

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

04

10 answers · Rationale-rich · Teach from doc

Q1**C**

C. $fx = 2331.70^{\frac{x}{2}}$

Choice C is correct. It's given that on April 1, there were 233 views of the advertisement. It's also given that every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. This situation can be represented by an exponential function of the form $fx = a1 + \frac{r}{100} \frac{x}{c}$, where a is the number of views on April 1 and every c days after April 1, the number of views had increased by $r\%$ of the number of views c days earlier. It follows that $a = 233$, $r = 70$, and $c = 2$. Substituting 233 for a , 70 for r , and 2 for c in the equation $fx = a1 + \frac{r}{100} \frac{x}{c}$ yields $fx = 2331 + \frac{70}{100} \frac{x}{2}$, or $fx = 2331.70^{\frac{x}{2}}$.

Choice A is incorrect. This function gives the predicted number of views for an advertisement for which every 2 days, the number of views was 70%, rather than increased by 70%, of the number of views 2 days earlier.

Choice B is incorrect. This function gives the predicted number of views for an advertisement for which every $\frac{1}{2}$ days, the number of views was 70% of the number of views $\frac{1}{2}$ days earlier, rather than an advertisement for which every 2 days, the number of views had increased by 70% of the number of views 2 days earlier.

Choice D is incorrect. This function gives the predicted number of views for an advertisement for which every $\frac{1}{2}$ days, rather than every 2 days, the number of views had increased by 70% of the number of views $\frac{1}{2}$ days earlier, rather than 2 days earlier.

Q2

B

B.

x	y
-6	-3
-5	-3
4	-3

Choice B is correct. It's given that $fx = (x + 6)(x + 5)(x - 4)$ and $y = f(x) - 3$. Substituting $(x + 6)(x + 5)(x - 4)$ for fx in the equation $y = f(x) - 3$ yields $y = (x + 6)(x + 5)(x - 4) - 3$. Substituting -6 for x in this equation yields $y = (-6 + 6)(-6 + 5)(-6 - 4) - 3$, or $y = -3$. Substituting -5 for x in the equation $y = (x + 6)(x + 5)(x - 4) - 3$ yields $y = (-5 + 6)(-5 + 5)(-5 - 4) - 3$, or $y = -3$. Substituting 4 for x in the equation $y = (x + 6)(x + 5)(x - 4) - 3$ yields $y = (4 + 6)(4 + 5)(4 - 4) - 3$, or $y = -3$. Therefore, when $x = -6$ then $y = -3$, when $x = -5$ then $y = -3$, and when $x = 4$ then $y = -3$. Thus, the table of values in choice B represents $y = f(x) - 3$.

Choice A is incorrect. This table represents $y = x - 3$ rather than $y = f(x) - 3$.

Choice C is incorrect. This table represents $y = x + 3$ rather than $y = f(x) - 3$.

Choice D is incorrect. This table represents $y = fx + 3$ rather than $y = f(x) - 3$.

Q3**D**

- D. For every increase in depth by 100 meters, the number of brachiopod species decreases by 10%.
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Choice D is correct. The function N is exponential, so it follows that $N(d)$ changes by a fixed percentage for each increase in d by 1. Since d is measured in hundreds of meters, it also follows that the number of brachiopod species changes by a fixed percentage for each increase in ocean depth by 100 meters. Since the base of the exponent in the model is 0.90, which is less than 1, the number of brachiopod species decreases as the ocean depth increases. Specifically, the number of brachiopod species at a depth of $d + 100$ meters is 90% of the number of brachiopod species at a depth of d meters. This means that for each increase in ocean depth by 100 meters, the number of brachiopod species decreases by 10%.

Choices A and C are incorrect. These describe situations where the number of brachiopod species are decreasing linearly rather than exponentially. Choice B is incorrect and results from interpreting the decrease in the number of brachiopod species as 10% for every 1-meter increase in ocean depth rather than for every 100-meter increase in ocean depth.

Q4**C**

- C. 1 day after the new product launched, the company's estimated stock price is \$ 83.
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Choice C is correct. In the given equation, x represents the number of days after a new product launched, where $0 \leq x \leq 20$, and y represents the estimated stock price, in dollars, for a certain company. Therefore, the best interpretation of $x, y = 1, 83$ in this context is that 1 day after the new product launched, the company's estimated stock price is \$ 83.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q5**B****B.** $\frac{1}{4}$

Choice B is correct. It's given that the formula $P = C2^{rt}$ gives the number of bacteria in a growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. It's also given that a scientist initially measures 12,000 bacteria in the growth medium. Since the initial measurement is 0 hours after the initial measurement, it follows that when $t = 0$, $P = 12,000$. Substituting 0 for t and 12,000 for P in the given equation yields $12,000 = C2^{r0}$, or $12,000 = C2^0$, which is equivalent to $12,000 = C$. It's given that 4 hours later, the scientist measures 24,000 bacteria, or when $t = 4$, $P = 24,000$. Substituting 4 for t , 24,000 for P , and 12,000 for C in the given equation yields $24,000 = 12,0002^{4r}$. Dividing each side of this equation by 12,000 yields $2 = 2^{4r}$, or $2^1 = 2^{4r}$, which is equivalent to $1 = 4r$. Dividing both sides of this equation by 4 yields $\frac{1}{4} = r$. Therefore, the value of r is $\frac{1}{4}$.

Choice A is incorrect. This is the value of the reciprocal of C .

Choice C is incorrect. This is the value of the reciprocal of r .

Choice D is incorrect. This is the value of C .

Q6**B****B.** $fx = 331.5^x$

Choice B is correct. Each of the given choices is an equation of the form $fx = ab^{x-k}$, where a , b , and k are constants. For an equation of this form, the coefficient, a , is equal to the value of the function when the exponent is equal to 0, or when $x = k$. It follows that in the equation $fx = 331.5^x$, the coefficient, 33, is equal to the value of $f0$. Substituting 0 for x in this equation yields $f0 = 331.5^0$, which is equivalent to $f0 = 331$, or $f0 = 33$. Thus, the value of c is 33 and the equation $fx = 331.5^x$ shows the value of c as the coefficient.

Choice A is incorrect. This equation shows the value of $f-1$, not $f0$, as the coefficient.

Choice C is incorrect. This equation shows the value of $f1$, not $f0$, as the coefficient.

Choice D is incorrect. This equation shows the value of $f2$, not $f0$, as the coefficient.

Q7**C****C. 7**

Choice C is correct. Substituting -1 for x in the given function f gives

$f(-1) = (-1)^3 + 3(-1)^2 - 6(-1) - 1$, which simplifies to $f(-1) = -1 + 3(1) - 6(-1) - 1$.

This further simplifies to $f(-1) = -1 + 3 + 6 - 1$, or $f(-1) = 7$.

Choice A is incorrect and may result from correctly substituting -1 for x in the function but incorrectly simplifying the resulting expression to $f(-1) = -1 - 3 - 6 - 1$, or -11 . Choice B is incorrect and may result from arithmetic errors. Choice D is incorrect and may result from correctly substituting -1 for x in the function but incorrectly simplifying the expression to $f(-1) = 1 + 3 + 6 + 1$, or 11 .

Q8**C**

C. $px = 2,0001.03^x$

Choice C is correct. It's given that a function p estimates that there were 2,000 animals in a population in 1998 and that each year from 1998 to 2010, the number of animals in this population increased by 3% of the number of animals in the population the previous year. It follows that this situation can be represented by the function $px = a1 + \frac{r}{100}^x$, where px is the estimated number of animals in the population x years after 1998, a is the estimated number of animals in the population in 1998, and each year the estimated number of animals increased by $r\%$. Substituting 2,000 for a and 3 for r in this function yields $px = 2,0001 + \frac{3}{100}^x$, or $px = 2,0001.03^x$.

Choice A is incorrect. This function represents a population in which each year the number of animals increased by 200%, not 3%, of the number of animals in the population the previous year.

Choice B is incorrect. This function represents a population in which each year the number of animals increased by 97%, not 3%, of the number of animals in the population the previous year.

Choice D is incorrect. This function represents a population in which each year the number of animals decreased, rather than increased, by 3% of the number of animals in the population the previous year.

Q9**D**

D. $y = 44,0002^t$

Choice D is correct. Since the number of bacteria doubles every day, the relationship between t and y can be represented by an exponential equation of the form $y = ab^t$, where a is the number of bacteria at the start of the observation and the number of bacteria increases by a factor of b every day. It's given that there are 44,000 bacteria at the start of the observation. Therefore, $a = 44,000$. It's also given that the number of bacteria doubles, or increases by a factor of 2, every day. Therefore, $b = 2$. Substituting 44,000 for a and 2 for b in the equation $y = ab^t$ yields $y = 44,0002^t$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the number of bacteria is decreasing by half, not doubling, every day.

Q10**B**

B. A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.

Choice B is correct. It's given that the graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole. It follows that for the point $(149.02, 186.05)$, 149.02 represents the molecular weight, in grams per mole, of a normal paraffin and 186.05 represents its estimated boiling point, in degrees Celsius. Therefore, the best interpretation of the point $(149.02, 186.05)$ is that a normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.

Choice A is incorrect. This statement represents a situation where x , not y , represents the boiling point, in degrees Celsius, and y , not x , represents the molecular weight, in grams, per mole.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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